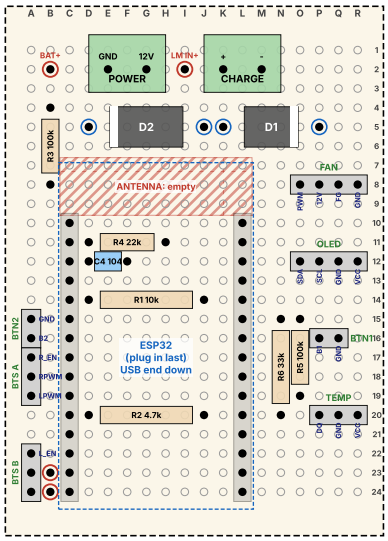
ThermoX UNIVERSAL PCB - TRUE SIZE (1:1) [ESP32 rows 22.86 mm]

Print at 100 % / Actual size (NOT 'fit to page'). Check the 50 mm bar with a ruler before you cut.

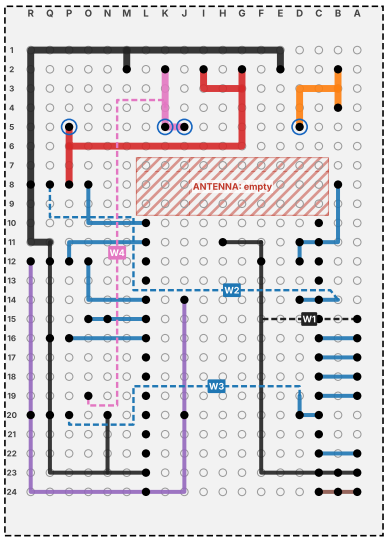
A. PARTS SIDE (paste this one)



BTS B: A22 L_EN, A23 GND, A24 VCC. Red rings: B23 = LM2596 OUT-, B24 = LM2596 OUT+

50 mm (A1 to R1 = 43.18 mm, A1 to A24 = 58.42 mm)

B. SOLDER SIDE (keep next to you)



Seen from the bottom: column A is on the RIGHT

HOW TO USE THIS PAGE

1. Print this page at 100 % (Actual size). Measure the 50 mm bar. If it is not 50 mm, fix the printer setting.
2. Cut copy A along the dashed line. Lay it on the PARTS side of the 5 × 7 cm board, letters A-R at the top.
3. Push a needle through the corner holes A1, R1, A24 and R24 into the board holes so every printed hole sits on a real hole. Tape the paper down (thin double-sided tape at the edges).
4. Push each part through the paper where it is drawn: low parts first (R1, R2, R4, C4 under the ESP32), then R3, R5, R6, D1, D2 (white band = stripe), the 2 female headers, the male headers, the 2 screw terminals.
5. Turn the board over. Copy B shows the same holes seen from the bottom: make every coloured line with bare tinned wire / solder from hole to hole, and the dashed W1-W4 with thin insulated wire.
6. Use pages 2-4 (big drawings and the hole-by-hole checklist) and tick each line when it is done.

black dot:	a part leg / pin goes in this hole
red ring:	solder a wire here (no part)
blue ring:	open this hole with a 1.5 mm drill (thick 1N5822 legs)
white band:	diode stripe (cathode) side
dashed blue:	ESP32 outline: it sits on the two grey female headers

TWO CAPACITORS:

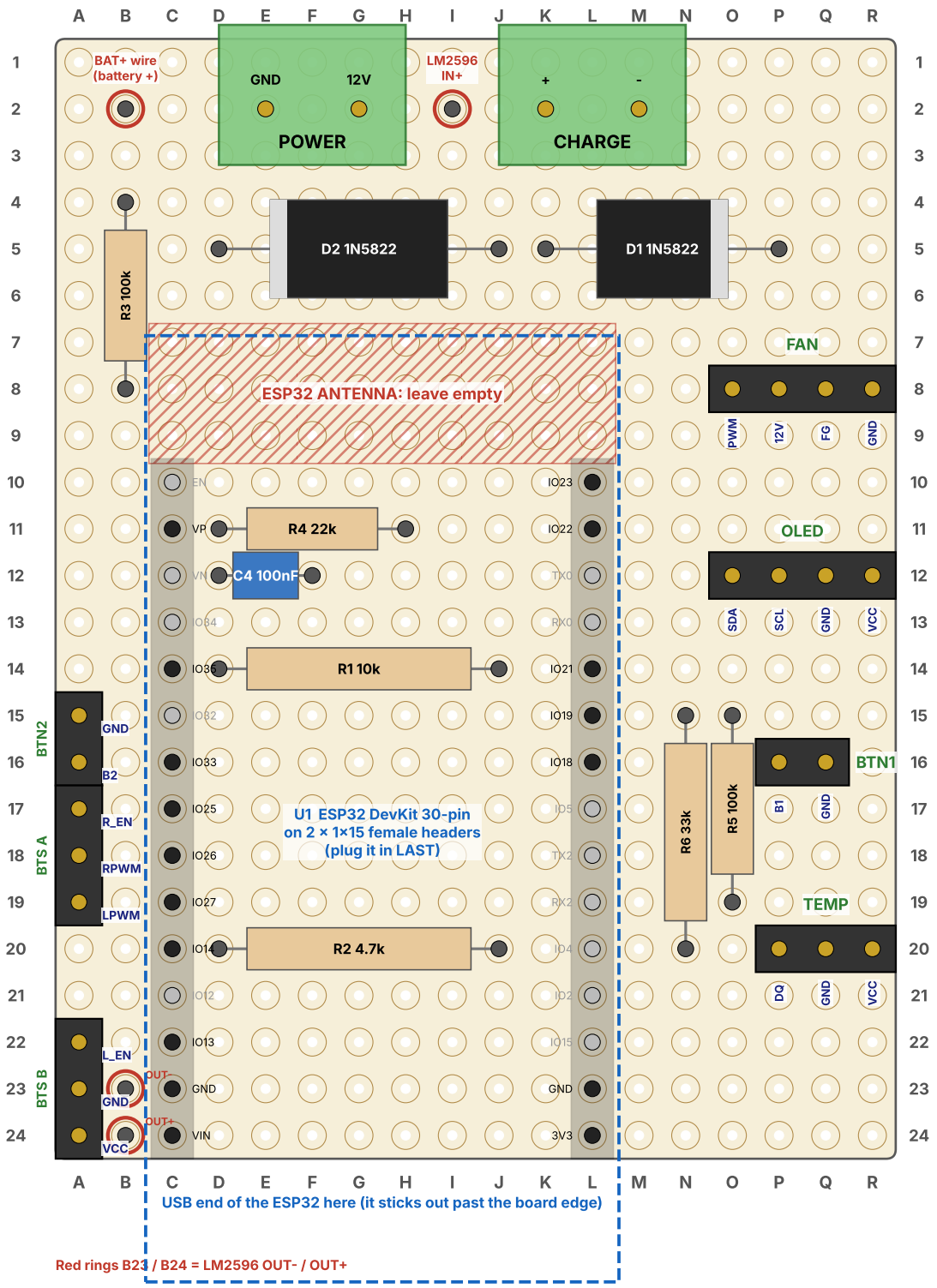
C4 = 100 nF ceramic (marked 104), ON this board in D12 and F12 (blue, under the ESP32). No + or -, any direction.

C1 = 2200 uF 25 V electrolytic, NOT on this board: screw it into the BTS7960 B+ and B- terminals, stripe (-) to B-. See the wiring diagram, page 6.

ground | 12 V | battery + | charger + | 5 V | 3.3 V | signals

ThermoX UNIVERSAL PCB - 2. PARTS SIDE, ZOOM (not to scale) [ESP32 rows 22.86 mm]

5 × 7 cm perfboard, 18 × 24 holes. Columns A-R, rows 1-24. Mark A, R, 1 and 24 on the board with a marker.



Gold dots = header / terminal pins. Grey dots = part legs. Red rings = wire pads (no part, a wire is soldered here).

R1, R2, R4, C4 lie FLAT under the ESP32 (between the two header rows): solder them before the female headers.

D1, D2 (1N5822): the grey band (stripe) is the cathode. D2 stripe at D5, D1 stripe at P5. Their legs are thick:
open holes D5, J5, K5, P5 with a 1.5 mm drill bit.

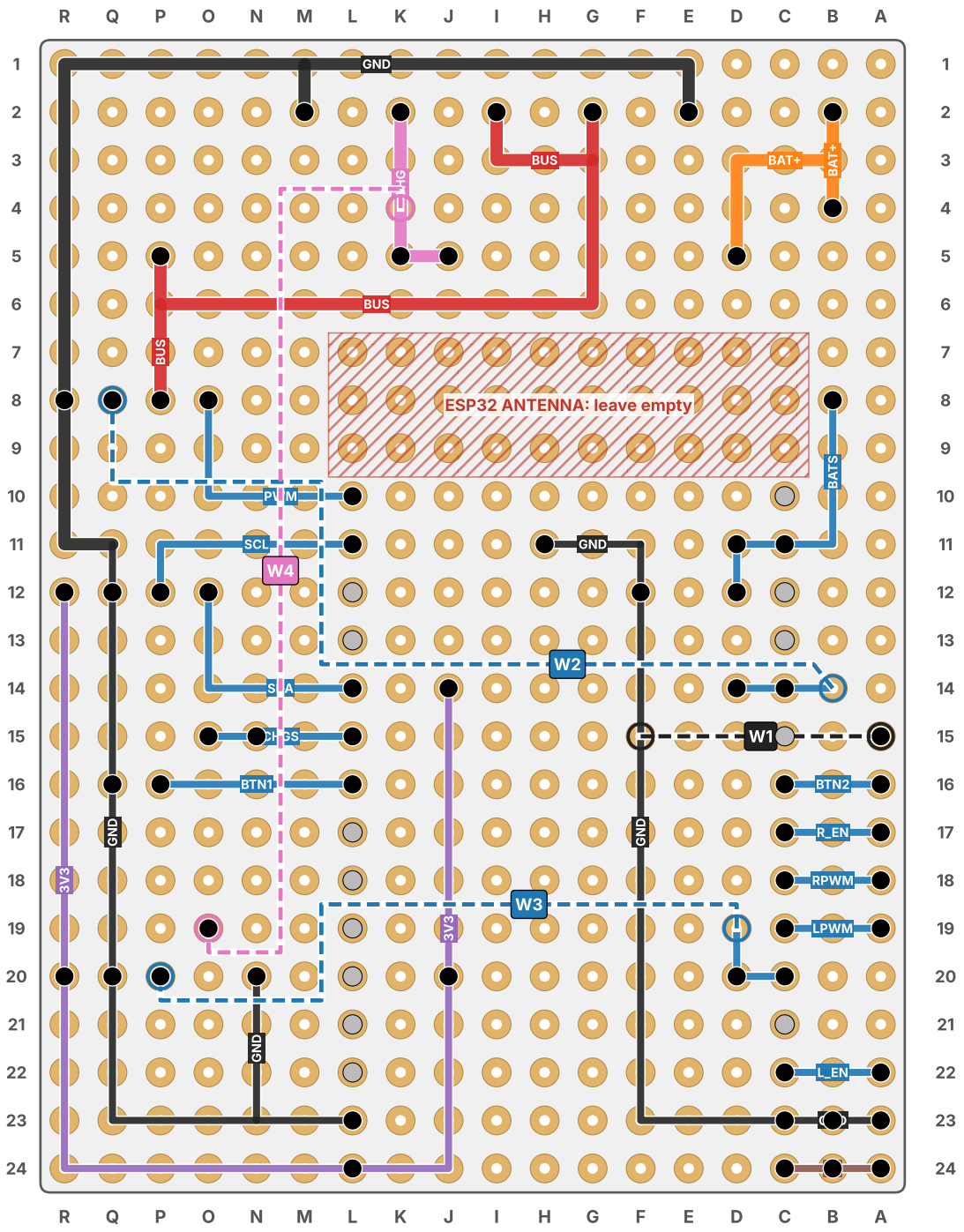
R5 and R6 lie flat side by side: R6 from N15 to N20, R5 from O15 to O19.

FAN, OLED, TEMP and BTN1 headers lie ACROSS (left to right). Read the pin names under each pin.

Nothing under the ESP32 antenna (hatched area): no parts and no trails.

ThermoX UNIVERSAL PCB - 3. SOLDER SIDE, ZOOM (not to scale) [ESP32 rows 22.86 mm]

Board turned over left to right: column A is now on the RIGHT, row 1 still at the top.



GND: ground
BUS: 12 V after the fuse
BAT+: battery + (gauge, charging)
CHG: charger +
5V: 5 V from the LM2596
3V3: 3.3 V from the ESP32
SIG: signals

Thick lines: bare tinned copper wire 0.8-1 mm (12 V, battery +, charger, top ground). Thin lines: a solder bridge or a bare wire from hole to hole. Black dots are part legs: the trail is soldered to them.

Dashed W1-W4: insulated wire (0.5 mm2 or thinner). They may cross the bare trails. Lay them flat.

Holes next to each other that belong to different colours must NOT touch. Check every joint with a meter.

ThermoX UNIVERSAL PCB - 4. BUILD CHECKLIST [ESP32 rows 22.86 mm]

A. PREPARE

[ ] Use a 5 × 7 cm board (18 × 24 holes). This PDF is for ESP32 pin rows 22.86 mm apart: right header in column L.

[ ] Write A ... R along the top edge and 1 ... 24 down the left edge, on BOTH sides (on the bottom A is on the right).

[ ] Open D5, J5, K5 and P5 with a 1.5 mm drill bit (the 1N5822 legs are thick).

B. PARTS ON TOP (low parts first). Push each leg through, bend it a little, solder, cut.

Step	Part	Holes	Note
[ ] 1	R1 10k	D14 to J14	flat, under the ESP32
[ ] 2	R2 4.7k	D20 to J20	flat, under the ESP32
[ ] 3	R4 22k	D11 to H11	flat, under the ESP32
[ ] 4	C4 100 nF	D12 and F12	under the ESP32, keep it low
[ ] 5	R3 100k	B4 to B8	flat, along column B
[ ] 6	R5 100k	O15 to O19	flat, along column O
[ ] 7	R6 33k	N15 to N20	flat, along column N
[ ] 8	D2 1N5822	J5 to D5	stripe at D5 (open the holes to 1.5 mm)
[ ] 9	D1 1N5822	K5 to P5	stripe at P5 (open the holes to 1.5 mm)
[ ] 10	ESP32 female headers 1x15	C10-C24 and L10-L24	plug the ESP32 in them to align, then solder
[ ] 11	BTN2 2-pin header	A15 (GND), A16 (B2)	
[ ] 12	BTS A 3-pin header	A17 R_EN, A18 RPWM, A19 LPWM	
[ ] 13	BTS B 3-pin header	A22 L_EN, A23 GND, A24 VCC (5 V)	
[ ] 14	FAN 4-pin header	O8 PWM, P8 12V, Q8 FG, R8 GND	lies across
[ ] 15	OLED 4-pin header	O12 SDA, P12 SCL, Q12 GND, R12 VCC	lies across
[ ] 16	BTN1 2-pin header	P16 (B1), Q16 (GND)	lies across
[ ] 17	TEMP 3-pin header	P20 DQ, Q20 GND, R20 VCC (3.3 V)	lies across
[ ] 18	POWER screw terminal 2-pin	E2 GND, G2 12V	wire openings toward the top edge
[ ] 19	CHARGE screw terminal 2-pin	K2 +, M2 -	wire openings toward the top edge

C. TRAILS ON THE BOTTOM (page 3). Follow each line hole by hole; '>' means a corner.

Net	From > corner > ... > to	Wire
[ ] GND	E2 > E1 > R1 > R11 > Q11	bare 0.8-1 mm
[ ] GND	M1 > M2	bare 0.8-1 mm
[ ] GND	A23 > C23	bare / solder
[ ] GND	C23 > F23 > F11	bare / solder
[ ] GND	F11 > H11	bare / solder
[ ] GND	L23 > Q23 > Q11	bare / solder
[ ] GND	N20 > N23	bare / solder
[ ] BUS	G2 > G3 > I3 > I2	bare 0.8-1 mm
[ ] BUS	G3 > G6 > P6 > P5	bare 0.8-1 mm
[ ] BUS	P6 > P8	bare 0.8-1 mm
[ ] CHG	K2 > K5	bare 0.8-1 mm
[ ] CHG	J5 > K5	bare 0.8-1 mm
[ ] BAT+	B2 > B4	bare 0.8-1 mm
[ ] BAT+	B3 > D3 > D5	bare 0.8-1 mm
[ ] 5V	A24 > C24	bare / solder
[ ] 3V3	L24 > J24 > J14	bare / solder
[ ] 3V3	L24 > R24 > R12	bare / solder
[ ] BATS	B8 > B11 > D11 > D12	bare / solder
[ ] FG	B14 > C14 > D14	bare / solder
[ ] DQ	C20 > D20 > D19	bare / solder
[ ] PWM	L10 > O10 > O8	bare / solder
[ ] SCL	L11 > P11 > P12	bare / solder
[ ] SDA	L14 > O14 > O12	bare / solder
[ ] CHGS	L15 > O15	bare / solder
[ ] BTN1	L16 > P16	bare / solder
[ ] BTN2	A16 > C16	bare / solder
[ ] R_EN	A17 > C17	bare / solder
[ ] RPWM	A18 > C18	bare / solder
[ ] LPWM	A19 > C19	bare / solder
[ ] L_EN	A22 > C22	bare / solder

ThermoX UNIVERSAL PCB - 5. WIRES, CHECKS, FIRST POWER-UP [ESP32 rows 22.86 mm]

D. INSULATED WIRES ON THE BOTTOM (dashed on page 3)

```
[ ] W1  GND  from A15  to F15   (button 2 GND)
[ ] W2  FG   from B14  to Q8    (fan tach)
[ ] W3  DQ   from D19  to P20   (temp sensor data)
[ ] W4  CHG  from K4   to O19   (charger detect)
```

Lay them flat along the path drawn on page 3, not across the ESP32 antenna end.

E. CHECK WITH A METER (beep mode), ESP32 NOT plugged in, nothing connected

```
[ ] Must NOT beep:  B2 (BAT+) to E2 (GND)   G2 (12V) to E2   K2 (CHG) to M2 (GND)   B24 (5V) to B23 (GND)
[ ] Must NOT beep:  R24 (3V3) to Q23 (GND)   B2 to G2   K2 to G2   each pin to its neighbours on the headers
[ ] Must beep:      E2 to M2, E2 to Q12, E2 to R8 (all GND)   G2 to I2, G2 to P8 (12V)   B2 to D5 (BAT+)
[ ] Must beep:      B24 to A24 and C24 (5V)   L24 to R20 and J14 (3V3)   K2 to O19 (CHG, via W4)
```

F. WIRING DIAGRAMS (pages 6-8, fuses on page 9), then FIRST POWER-UP without the ESP32

[ ] Set the LM2596 to 5.0 V with a meter BEFORE wiring it. Set the USB-C trigger board to 12 V and measure it.

[ ] Wire the LM2596: IN+ to I2, OUT+ to B24, OUT- to B23 (its IN- is the same as OUT-, no wire needed).

[ ] Battery on, switch on BATT: G2 = battery voltage, B24 = 5.0 V, B2 = battery voltage. Switch to OFF: G2 = 0 V.

[ ] Switch OFF, charger in the DC jack: K2 = about 12.6 V, G2 = about 12 V (D1 feeds the board), battery rises.

[ ] Everything OK: switch off, plug in the ESP32 (USB end at the board edge), connect the headers.

G. TEST AND USE

[ ] Upload ThermoX_Bench/ThermoX_Bench.ino, Serial Monitor 115200: check TEMP, BATT, CHARGER, B1/B2, fan (f), Peltier (h, c).

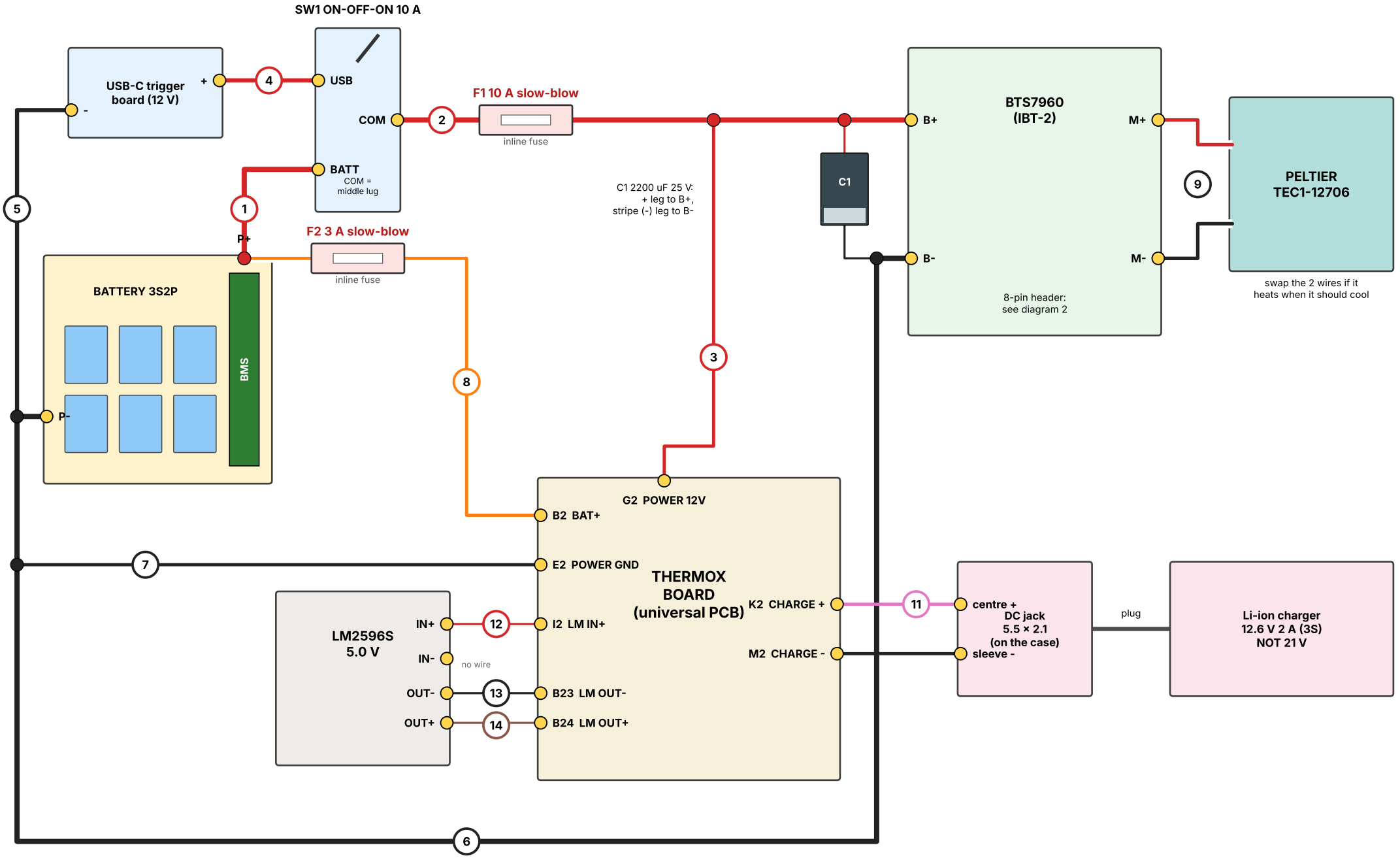
[ ] Upload ThermoX/ThermoX.ino. Set ENABLE_BATTERY_SENSE = true (the battery divider R3/R4 is on this board).
ENABLE_CHARGE_DETECT stays true (R5/R6 are on this board).

[ ] Pins used: Button 1 GPIO18, Button 2 GPIO33, fan PWM GPIO23, fan FG GPIO35, battery GPIO36 (VP),
charger GPIO19, DS18B20 GPIO14, OLED SDA 21 / SCL 22, BTS7960 R_EN 25, RPWM 26, LPWM 27, L_EN 13.

[ ] The DS18B20 runs from 3.3 V on this board (TEMP VCC = 3.3 V). That is fine for the DS18B20.

ThermoX WIRING DIAGRAM 1 - POWER [ESP32 rows 22.86 mm]

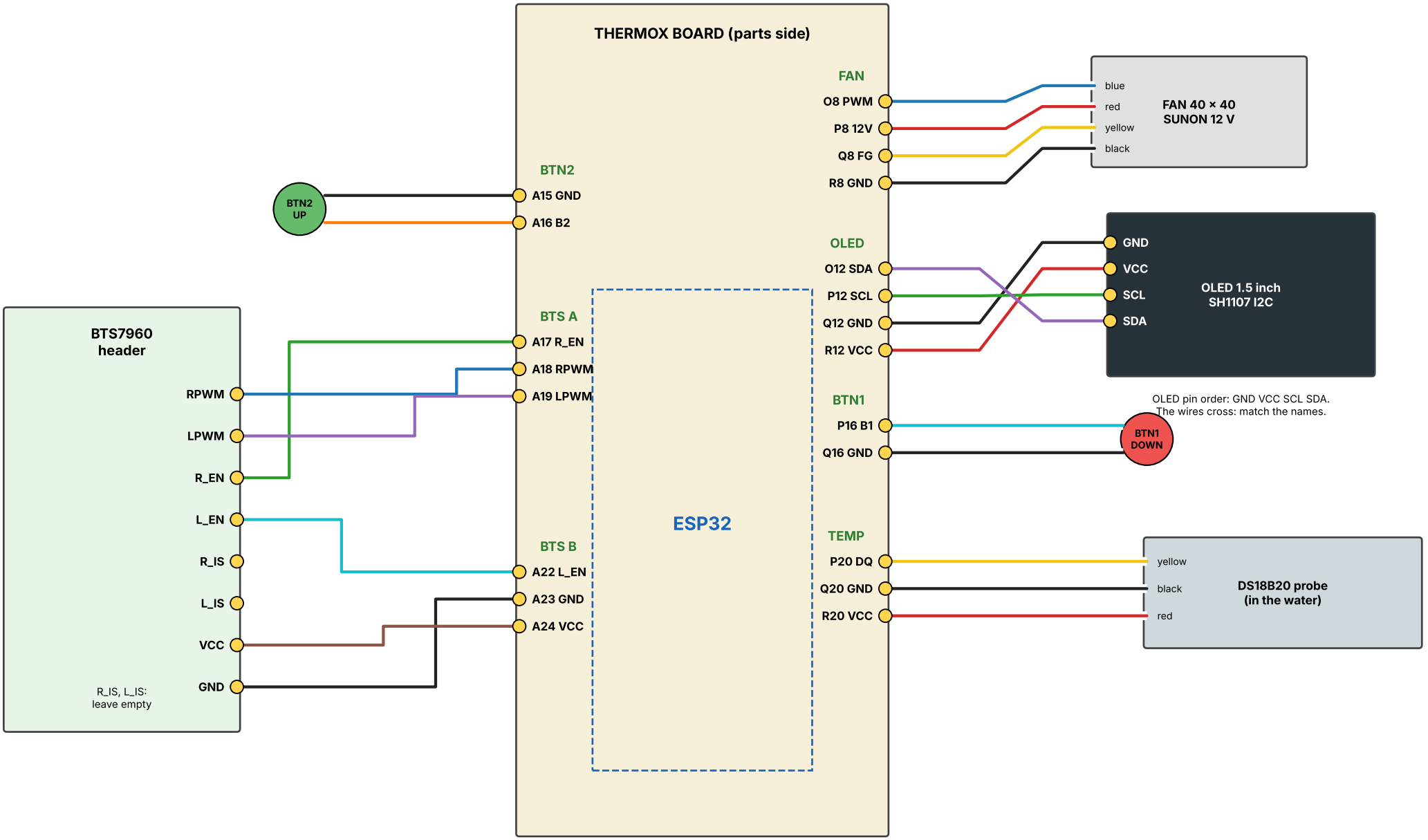
Thick lines = 1.5 mm2 (Peltier current). Dots = joined. Numbers = wire list on the next page. Lines that cross without a dot are NOT joined.



Before the battery goes on: SW1 in the middle (OFF), fuses in, LM2596 set to 5.0 V, trigger board set to 12 V. Never plug the USB-C charger and the DC charger in at the same time.

ThermoX WIRING DIAGRAM 2 - HEADERS TO THE PARTS (Dupont wires) [ESP32 rows 22.86 mm]

Each line is one female-female Dupont wire. Connect by NAME: the order of the pins on a module can be different from the board.



The BTS7960 B+, B-, M+ and M- screw terminals are on diagram 1 (thick wires). Short Dupont wires are best: a loose plug can stop the fan or the driver.

ThermoX WIRE LIST [ESP32 rows 22.86 mm]

Tick each wire when it is done.

POWER WIRES (diagram 1)

No.	From	To	Wire	Colour	Note
[ ] 1	Battery P+ (BMS)	SW1 BATT lug	1.5 mm2	red	
[ ] 2	SW1 COM (middle lug)	F1 10 A → BTS7960 B+	1.5 mm2	red	F1 inline in this wire
[ ] 3	Wire 2 after F1 (splice)	Board POWER 12V (G2)	0.75 mm2	red	solder + heat-shrink the splice
[ ] 4	Trigger board +	SW1 USB lug	1 mm2	red	
[ ] 5	Trigger board -	Battery P- (BMS)	1 mm2	black	Peltier current on USB-C
[ ] 6	Battery P- (BMS)	BTS7960 B-	1.5 mm2	black	
[ ] 7	Battery P- (BMS)	Board POWER GND (E2)	0.75 mm2	black	
[ ] 8	Battery P+ (BMS)	F2 3 A → Board BAT+ (B2)	0.75 mm2	orange	F2 within 5 cm of P+
[ ] 9	BTS7960 M+ and M-	Peltier red and black	Peltier wires	red/black	swap them if it heats when it should cool
[ ] 10	C1 2200 uF + leg / stripe (-) leg	BTS7960 B+ / B-	C1 legs	-	in the screws with wires 2 and 6
[ ] 11	DC jack centre (+) / sleeve (-)	Board CHARGE + (K2) / - (M2)	0.5-0.75 mm2	pink/black	
[ ] 12	LM2596 IN+	Board LM2596 IN+ (I2)	0.5 mm2	red	
[ ] 13	LM2596 OUT-	Board B23	0.5 mm2	black	LM2596 IN- needs no wire (same as OUT-)
[ ] 14	LM2596 OUT+	Board B24 (5 V)	0.5 mm2	brown	set the LM2596 to 5.0 V first

DUPONT WIRES (diagram 2)

```
[ ] FAN    O8   PWM   -> fan BLUE wire
[ ] FAN    P8   12V   -> fan RED wire
[ ] FAN    Q8   FG    -> fan YELLOW wire
[ ] FAN    R8   GND   -> fan BLACK wire
[ ] OLED   O12  SDA   -> OLED SDA
[ ] OLED   P12  SCL   -> OLED SCL
[ ] OLED   Q12  GND   -> OLED GND
```

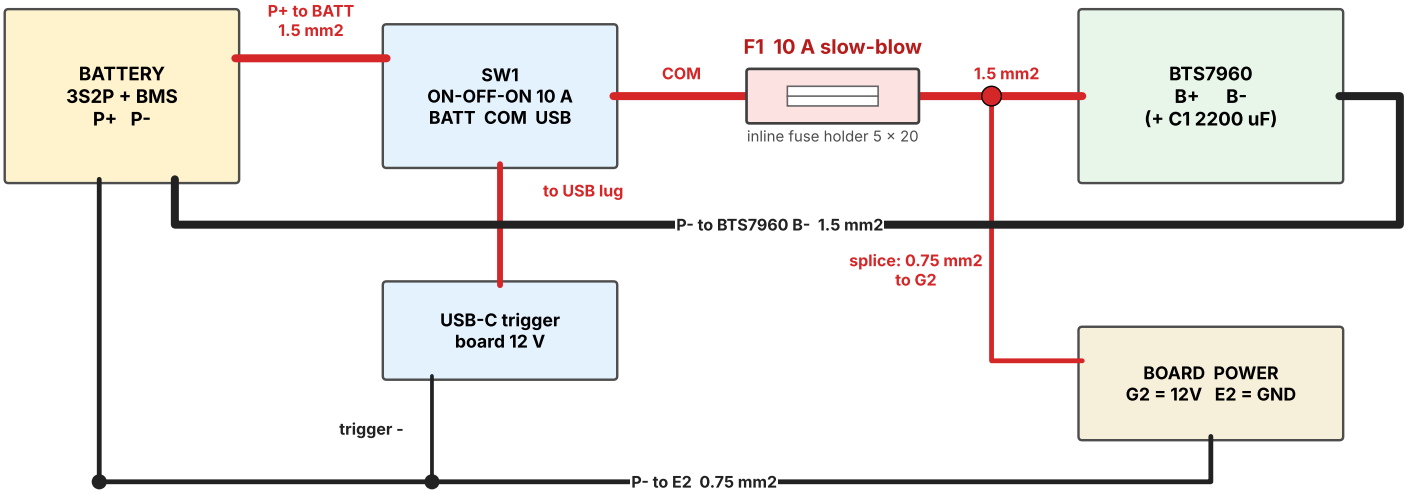
```
[ ] OLED   R12  VCC   -> OLED VCC (3.3 V)
[ ] BTN1   P16  B1    -> button 1 (DOWN), one leg
[ ] BTN1   Q16  GND   -> button 1, other leg
[ ] TEMP   P20  DQ    -> DS18B20 YELLOW (data)
[ ] TEMP   Q20  GND   -> DS18B20 BLACK
[ ] TEMP   R20  VCC   -> DS18B20 RED
[ ] BTN2   A15  GND   -> button 2, other leg
```

```
[ ] BTN2   A16  B2    -> button 2 (UP), one leg
[ ] BTS A  A17  R_EN  -> BTS7960 R_EN
[ ] BTS A  A18  RPWM  -> BTS7960 RPWM
[ ] BTS A  A19  LPWM  -> BTS7960 LPWM
[ ] BTS B  A22  L_EN  -> BTS7960 L_EN
[ ] BTS B  A23  GND   -> BTS7960 GND
[ ] BTS B  A24  VCC   -> BTS7960 VCC (5 V)
```

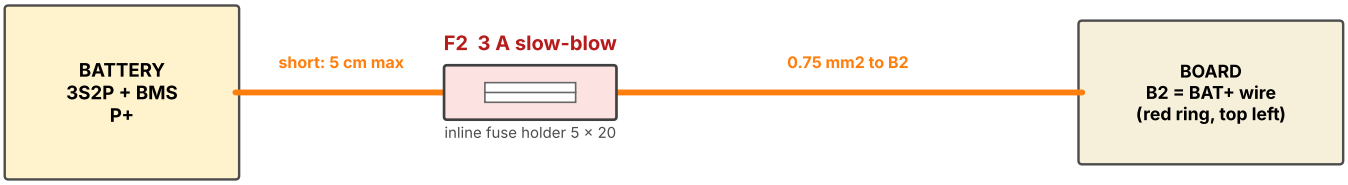
ThermoX - 9. FUSES AND FAN WIRES [ESP32 rows 22.86 mm]

The fuses are NOT on the board on purpose: the Peltier current (up to 7 A) must not go through a perfboard.

1. F1 10 A - main 12 V line (Peltier, fan, board)



2. F2 3 A - BAT+ wire to the board (battery gauge and charging)



HOW TO FIT EACH INLINE FUSE HOLDER

1. Battery disconnected (or BMS off) while you wire. SW1 in the middle (OFF).
2. Cut the wire where the fuse goes. Strip 8 mm of each end and of both holder leads.
3. Slide heat-shrink on first, then twist each end to one holder lead, solder, and shrink the tube over the joint.
4. F1: the splice to G2 is on the BTS7960 side of F1 (after the fuse). Solder the 0.75 mm2 wire to the 1.5 mm2 wire and cover it with heat-shrink. Both the BTS7960 and the board are then protected by F1.
5. F2: put it within 5 cm of the battery P+ so the whole thin wire to B2 is protected. Solder the F2 lead and the 1.5 mm2 SW1 wire both on the BMS P+ pad.
6. Put the fuses in: 10 A slow-blow in F1, 3 A slow-blow in F2. Never a bigger fuse, never a wire instead.
7. Check with a meter (fuses in, battery off): SW1 COM to G2 beeps (through F1); battery P+ to B2 beeps (through F2).
8. A fuse that blows means a short: find it before you put a new fuse in.

FAN: SUNON 40 × 40 × 28 mm, 12 V, 4 wires → FAN header O8-R8 (row 8, lies across)

BLUE → O8 PWM: speed control, GPIO23	**RED → P8** 12V: +12 V (after F1)	**YELLOW → Q8** FG: speed signal (rpm), GPIO35	**BLACK → R8** GND: ground

These are the usual SUNON 4-wire colours. Check the sticker or datasheet of YOUR fan first: some fans use other colours. If the fan only has 3 wires (no blue), it has no PWM: leave O8 empty, it runs full speed whenever 12 V is on. With only 2 wires there is no FG either: set FAN_MIN_RPM = 0 in ThermoX.ino (stall check off).
The fan needs about 1 A at full speed. Use the Dupont cable only if it is short; a loose plug makes the fan stall.

ThermoX - 10. COMPLETE PARTS LIST (everything to buy) [ESP32 rows 22.86 mm]

ON THE UNIVERSAL PCB

[ ]	**1**	Universal PCB 5 × 7 cm, single-sided, 18 × 24 holes, 2.54 mm	
[ ]	**1**	ESP32 DevKit V1, 30 pins (ESP-WROOM-32)	U1
[ ]	**2**	Female header 1 × 15, 2.54 mm	socket for the ESP32
[ ]	**1**	Male header strip 1 × 40, 2.54 mm (cut into 4+4+3+3+3+2+2 = 21 pins)	FAN OLED TEMP BTS-A BTS-B BTN1 BTN2
[ ]	**2**	Screw terminal 2-pin 5.08 mm (KF301-2P)	POWER, CHARGE
[ ]	**1**	Resistor 10 kOhm 1/4 W	R1
[ ]	**1**	Resistor 4.7 kOhm 1/4 W	R2
[ ]	**2**	Resistor 100 kOhm 1/4 W	R3, R5
[ ]	**1**	Resistor 22 kOhm 1/4 W	R4
[ ]	**1**	Resistor 33 kOhm 1/4 W	R6
[ ]	**1**	Ceramic capacitor 100 nF (104)	C4
[ ]	**2**	Schottky diode 1N5822 (3 A)	D1, D2
[ ]	**1 m**	Bare tinned copper wire 0.8-1 mm	trails
[ ]	**1 m**	Thin insulated wire (0.25-0.5 mm2)	W1-W4

MODULES AND PARTS AROUND THE BOARD

[ ]	**1**	OLED 1.5 inch 128 × 128, SH1107, I2C (GME128128-01-IIC)	OLED header
[ ]	**1**	DS18B20 waterproof temperature probe (3 wires)	TEMP header
[ ]	**1**	BTS7960 (IBT-2) 43 A motor driver module	BTS A / BTS B headers
[ ]	**1**	Peltier TEC1-12706 (12 V, about 6 A)	BTS7960 M+ / M-
[ ]	**1**	SUNON fan 40 × 40 × 28 mm, 12 V, 4-wire (PWM + FG)	FAN header
[ ]	**1**	LM2596S DC-DC step-down module (set to 5.0 V)	I2, B23, B24
[ ]	**1**	USB-C PD trigger board 100 W 5 A (ZY12PDN), set to 12 V	SW1 USB
[ ]	**2**	Push button, momentary, panel mount	BTN1, BTN2
[ ]	**1**	ON-OFF-ON toggle switch, 10 A or more (MTS-103 is only 6 A)	SW1
[ ]	**2**	Inline fuse holder 5 × 20 mm (with wire leads)	F1, F2
[ ]	**2+2**	Fuse 5 × 20 mm slow-blow: 10 A (F1) and 3 A (F2), plus 1 spare of each	F1, F2
[ ]	**1**	Electrolytic capacitor 2200 uF 25 V low ESR	C1 at the BTS7960
[ ]	**1**	DC jack 5.5 × 2.1 mm, female, panel mount	charging input

BATTERY AND CHARGING

[ ]	**6**	Li-ion 21700 cell 3.7 V 5000 mAh, 40 A (Enook)	3S2P pack
[ ]	**1**	3S BMS protection board, 20 A, with balance wires	
[ ]	**1**	3S2P 21700 holder, or nickel strip 0.15 × 8 mm (spot welded)	builds the pack
[ ]	**1**	Li-ion charger 12.6 V 2 A (3S, CC/CV), 5.5 × 2.1 mm plug. NOT 21 V	
[ ]	**1**	USB-C PD charger that gives 12 V (60 W or more) + 5 A USB-C cable	optional, all-day use

WIRES AND SMALL PARTS

[ ]	**1 m + 1 m**	Wire 1.5 mm2 (16 AWG), red and black	battery, SW1, F1, BTS7960 B+/B-, Peltier
[ ]	**1 m + 1 m**	Wire 0.75 mm2 (18 AWG), red and black	G2, E2, B2, trigger board, charge jack
[ ]	**1 m**	Wire 0.5 mm2 (20 AWG)	LM2596
[ ]	**1 pack**	Dupont jumper wires female-female, 20 cm (40 pcs)	all headers
[ ]	**1 set**	Heat-shrink tubing, assorted	
[ ]		Solder 0.8 mm with flux, double-sided tape	

THERMAL AND BODY

[ ]	**1**	Copper heatsink for the Peltier hot side (fits the 40 mm fan)	
[ ]	**1**	Aluminium water cup (Peltier cold side on its floor)	
[ ]	**1**	Thermal paste (1 tube)	Peltier, cup, heatsink
[ ]	**4**	M3 screws + nuts or standoffs	board, fan
[ ]	**1**	Bottle body (3D printed, design from cad/)	

TOOLS

[ ]		Soldering iron, multimeter (beep + volts), wire stripper, side cutter
[ ]		Drill bits 1.5 mm (diode holes) and 1.3 mm (only if the screw terminal pins do not fit)